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CS 499

Professional Self-Assessment

Throughout my time in the Computer Science program at SNHU, I have learned new skills, refined skills I already possessed, and developed a better understanding of computer science as a field, including the different career opportunities within it. Completing coursework throughout the program and developing my portfolio during this course has helped me better understand my strengths and areas for improvement, as well as how to apply my strengths toward my future career goals while continuing to grow professionally.

Because my career goal is to become a data analyst, an important part of my growth in the program specifically was the coursework in DAD 220. During this course, I learned how data can be used to solve business problems, I learned how to use SQL to retrieve information from large databases to answer business questions. Along with learning technical skills in this course, I also learned communication and professional skills to assist me with my journey into data analytics.

Even though most of my coursework was completed individually, I still developed essential communication and collaboration skills through discussion posts with fellow students, communicating with instructors whenever I needed help, and participating in assignments that were meant to teach me how to communicate with stakeholders and non-technical audiences. In my discussion posts, I would explain the week's course work from my perspective in my initial posts as well as seek out other students' perspectives by replying to their posts. I also regularly communicated with my instructors whenever I got stuck on something in my work or needed

additional clarification that I could not find in the reading materials. In different courses, some assignments also required me to put myself in the shoes of different users of a system. This helped me develop the necessary communication skills non-technical audiences might need to understand a system. I believe this skill will be the most useful for data analysis as a large part of the job involves communicating and collaborating with others.

Throughout the computer science program, I also gained an understanding of data structures and algorithms, software design, and databases. The biggest challenge for me, working in these specific categories, was software design as it is not my chosen interest. At first, I had no understanding of how to structure my program, and my goal became more focused on simply getting my code to run rather than organizing it in the most efficient way. I was grateful to be able to return to this category in this course because I was able to critically analyze my previous code and think about more efficient ways to structure it. The algorithms and database categories came more naturally to me as I had already had some experience in them, but they also came with their own challenges, as I had to re-evaluate methods I had previously learned and apply new approaches to solve problems more effectively. In addition to these areas, I also learned and applied security concepts to my work. Learning about the importance of security withing computer science made me aware of how important it is to develop secure code and consider potential threats to systems. This helped me be more cautious while developing as I was more concerned with keeping data protected.

Overall, my artifacts come together to show the progress I have made throughout my time at SNHU and how my skills have developed over time. I have two artifacts I am presenting.

Artifact one is a program that shows a 3-D scene made of complex objects. I believe it shows my skills obtained in software development and design and shows how I improved the structure of

my program by removing the repetition and making my code more scalable and maintainable.

Artifact two is a personal project clothing store inventory database I believe shows both my skills in algorithms and data structures as well as my skills in databases. I made this artifact into a more complex relational database as it was incredibly simplistic in its original form.

Altogether, it is a demonstration of my ability to use analytical thinking to solve problems as well as work with larger, more complex databases.